

RA-17: Linear Recovery and Its Topological Relaxation

Sidney Holden

Center for Computational Biology
Flatiron Institute, Simons Foundation

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Abstract

This is a continuation of the two preceding RA-17 proof packages, not a complete classification of the minimum number of linear measurements. The main additional result is an exact solution of a topological relaxation: one measurement below the complex generic count, a continuous odd nonvanishing map exists if and only if the complex determinantal degree is even. The corresponding evaluation bundle also admits the required number of independent sections exactly in the even-degree case. These statements explain a genuine limit of the earlier bundle strategy, rather than an unevaluated characteristic number. A direct rank-one index calculation recovers the lower bound of nineteen measurements in dimension six and shows precisely why it does not exclude nineteen. Integer measurement constructions, rational arithmetic checks, and a separately labelled audit of the interrupted small-pencil calculation accompany the proofs. The existence or impossibility of a seventeen-dimensional real 6×6 minimum-rank-three pencil remains undetermined here. Consequently the overall status is **PARTIAL**.

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1 The exact target and the scope of this continuation

For integers $1 \leq r$ and $2r < d$, write $\mu_{\mathbb{R}}(d, r)$ for the smallest m for which a linear map

$$\mathcal{A} : M_d(\mathbb{R}) \longrightarrow \mathbb{R}^m$$

is injective on all matrices of rank at most r . This is the all-parameter classification requested by RA-17.¹ Define

$$k = 2r, \quad c = d - k, \quad m_0 = d^2 - c^2 = 4r(d - r), \quad n = m_0 - 1.$$

The dimension n is odd. The complex projective variety of matrices of rank at most k has dimension n and degree

$$D_{d,r} = \prod_{i=0}^{c-1} \frac{(d+i)! i!}{(k+i)! (c+i)!}. \quad (1)$$

This is the standard square determinantal degree; it is also used in the real recovery literature.²

The distinction between three questions is essential. The first asks for a *linear* $\mathcal{A}$ that does not vanish at any nonzero rank-at-most- k matrix. The second asks for a continuous odd nonvanishing map on the same unit link. The third asks whether the evaluation bundle has a specified number of independent sections. A positive answer to the first implies the other two. Neither converse is proved here. In particular, arbitrary sections of an evaluation bundle need not be restrictions of a fixed space of constant matrices.

The preceding package gives, among other results, the intervals

$$19 \leq \mu_{\mathbb{R}}(6, 1) \leq 20, \quad 35 \leq \mu_{\mathbb{R}}(10, 1) \leq 36.$$

The present argument independently checks the first lower bound in a short specialized calculation. It does not improve these intervals. The earlier archives are retained unchanged as provenance; retaining them does not mean that every earlier topological argument has been independently formalized or peer-reviewed. No claim of historical priority is made for the results below.

2 Linear kernels and integer upper-bound constructions

Lemma 2.1 (Kernel criterion). *Let $K = \ker \mathcal{A}$. Uniform injectivity on rank-at-most- r matrices is equivalent to*

$$A \in K \setminus \{0\} \implies \text{rank } A \geq k + 1.$$

Consequently, if $\kappa(d, r)$ is the maximum dimension of such a real linear space, then $\mu_{\mathbb{R}}(d, r) = d^2 - \kappa(d, r)$.

¹See source [1].

²See source [2].

Proof. Rank subadditivity puts every difference of two rank-at-most- r matrices in the rank-at-most- $2r$ set. Conversely, express a rank- ℓ matrix, $\ell \leq 2r$, as a sum of ℓ rank-one matrices, split the summands into two groups of size at most r , and put a minus sign into the second group. This expresses the matrix as the difference of two admissible signals. A nonzero such matrix in the kernel is therefore exactly a collision. Redundant measurement rows can be removed, and a basis of the orthogonal complement of any admissible kernel supplies $d^2 - \dim K$ measurements. $\square$

Proposition 2.2 (Explicit integer measurements). *For every admissible (d, r) , there is an integer-valued measurement system with $m_0 = 4r(d - r)$ independent rows satisfying Lemma 2.1.*

Proof. Treat each anti-diagonal separately. If its length is ℓ , enumerate its entries by $j = 1, \dots, \ell$ and measure their moments

$$\sum_{j=1}^{\ell} j^a X_j, \quad a = 0, \dots, \min(k, \ell) - 1.$$

The Vandermonde matrix has full row rank. If a vector in its kernel had at most k nonzero coordinates, the corresponding Vandermonde columns would be independent, so that vector would be zero. Thus a nonzero kernel vector on an anti-diagonal has at least $k + 1$ nonzero entries.

Take the first nonzero anti-diagonal of a nonzero full kernel matrix and select $k + 1$ nonzero positions. In increasing row order and decreasing column order, the resulting square submatrix is triangular: entries on one side of its selected diagonal belong to earlier, zero anti-diagonals. Its determinant is, up to sign, the product of the selected nonzero entries. Hence the matrix has rank at least $k + 1$. Finally,

$$\dim K = \sum_{\text{anti-diagonals}} \max(\ell - k, 0) = (d - k)^2 = c^2.$$

The measurements have disjoint anti-diagonal supports, so their full row rank follows from the rank of the individual Vandermonde blocks. $\square$

The exported file `data/antidiagonal_6_1.json` contains twenty explicit integer measurement rows and a rational basis of their sixteen-dimensional kernel. This is the certified upper endpoint, not a claimed nineteen-measurement construction. Other exported examples use exactly the same proof.

Proposition 2.3 (Odd degree gives the exact generic count). *If $D_{d,r}$ is odd, then $\mu_{\mathbb{R}}(d, r) = m_0$.*

Proof. Suppose a kernel of dimension at least $c^2 + 1$ avoided the real rank-at-most- k cone. Restrict to a $(c^2 + 1)$ -dimensional subspace. Avoidance is open on the real Grassmannian: the unit rank-at-most- k link is compact, and its distance from the projectivized kernel is positive. A sufficiently small generic real perturbation would therefore still avoid real points. But its complex projectivization meets the complex determinantal variety transversely in $D_{d,r}$ points. Nonreal points occur in conjugate pairs; odd degree forces a real point. This is a contradiction. Proposition 2.2 supplies equality. $\square$

3 An exact solution of the continuous odd relaxation

Let

$$X = \{[A] \in \mathbb{RP}^{d^2-1} : \text{rank } A \leq k\}, \quad Z = \{A : \text{rank } A \leq k, \|A\|_F = 1\}.$$

Let λ be the tautological real line bundle on X , and write $a = w_1(\lambda)$. The map $Z \rightarrow X$ is its two-sheeted sphere bundle.

Theorem 3.1 (Critical continuous relaxation). *There exists a continuous map*

$$f : Z \longrightarrow S^{n-1}, \quad f(-A) = -f(A),$$

if and only if $D_{d,r}$ is even. Equivalently, the bundle $n\lambda$ over X has a nowhere-zero section if and only if $D_{d,r}$ is even.

This assertion is at the critical count $n = m_0 - 1$. It does not purport to find the smallest count for the continuous relaxation, and it does not assert that f is the normalization of a linear map.

3.1 Orientability and the top local-coefficient group

Lemma 3.2. *The smooth rank- k stratum $U \subset X$ is connected and orientable. The restriction of λ to U has nontrivial sign monodromy. If $\mathbb{Z}_\lambda$ denotes the corresponding sign local system, then*

$$H^n(X; \mathbb{Z}_\lambda) \cong \mathbb{Z}/2, \quad H^n(X; \mathbb{F}_2) \cong \mathbb{F}_2,$$

and reduction modulo two between these top cohomology groups is an isomorphism.

Proof. Let $G = \text{Gr}_k(\mathbb{R}^d)$ and let S be its tautological rank- k bundle. The projective bundle

$$\pi : P(dS) \longrightarrow G$$

resolves X by recording the image plane of a matrix. It is a diffeomorphism over U . Put $u = w_1(S)$ and $h = w_1(\lambda)$ on this projective bundle. The standard tangent-bundle formulas give

$$w_1(TG) = du, \quad w_1(dS) = du, \quad w_1(TP(dS)) = dk h + \pi^*(du + du) = 0.$$

Here dk is even because $k = 2r$. Thus the resolution, and its open subset U , are orientable. These formulas follow from $TG \cong \text{Hom}(S, S^\perp)$ and $T_{\text{vert}}P(E) \oplus \varepsilon \cong \lambda^* \otimes \pi^*E$.³

The full-rank $k \times d$ matrices form a connected space when $k < d$: orthonormal row frames are connected in this range, and the positive-definite factor in a polar decomposition is connected. Together with connectedness of G , this proves connectedness of U . Since k is even, $-I_k \in SO(k)$ can be joined to I_k by a path. Applying that path on the image of a fixed rank- k matrix joins A to $-A$ without reducing the rank. Projectivization produces a loop on which λ reverses sign.

The singular subset $Y = X \setminus U$ has codimension

$$\dim X - \dim Y = 2(d - k) + 1 = 2c + 1 \geq 3.$$

Compact semialgebraic sets admit finite triangulations.⁴ The long exact sequence for (X, Y) therefore identifies top cohomology of X with compactly supported top cohomology of U , with either coefficient system. Poincare duality on the connected orientable manifold U gives

$$H_c^n(U; \mathbb{Z}_\lambda) \cong H_0(U; \mathbb{Z}_\lambda) = \mathbb{Z}/2.$$

The last equality is the coinvariant calculation: transport around the loop above imposes $v = -v$. With mod-two coefficients the result is $\mathbb{F}_2$. The coefficient sequence $0 \rightarrow \mathbb{Z}_\lambda \xrightarrow{2} \mathbb{Z}_\lambda \rightarrow \mathbb{F}_2 \rightarrow 0$ now shows that reduction in degree n is an isomorphism; multiplication by two is zero and there is no cohomology above dimension n . $\square$

³See source [4].

⁴See source [6].

3.2 The Euler obstruction and degree parity

Proof of Theorem 3.1. A nowhere-zero section of the rank- n bundle $n\lambda$ is a section of its sphere bundle with fiber S^{n-1} . Since X is an n -dimensional finite CW space and S^{n-1} is $(n-2)$ -connected, its only obstruction is the Euler class in degree n . This is obstruction theory for vector-bundle sections, with the orientation local system included.⁵

Because n is odd, that orientation system is $\mathbb{Z}_\lambda$. The Euler class reduces to

$$w_n(n\lambda) = a^n.$$

By Lemma 3.2, the Euler class is zero exactly when a^n is zero. Intersecting X with n generic real projective hyperplanes computes the pairing of a^n with its mod-two fundamental class. The intersection avoids Y and is transverse. Its complexification consists of $D_{d,r}$ points, so its real cardinality is congruent to $D_{d,r}$ modulo two. Since top mod-two cohomology is one-dimensional, $a^n = 0$ exactly when $D_{d,r}$ is even.

Finally a section of $n\lambda$ pulls back to Z in the form $f(A) \otimes A$. Independence of the choice of A or $-A$ is precisely $f(-A) = -f(A)$. A nowhere-zero section can be normalized to have $\|f(A)\| = 1$. This proves both equivalences. $\square$

Corollary 3.3 (Polynomial, but not necessarily linear, realization). *When $D_{d,r}$ is even, there exist n rational homogeneous polynomials of one common odd degree with no common nonzero zero on the rank-at-most- k cone. No bound of one on that degree is asserted.*

Proof. Approximate the coordinates of the map in Theorem 3.1 uniformly on compact Z by polynomial restrictions. Polynomials separate points, so Stone–Weierstrass applies. Replace a polynomial $p(A)$ by $(p(A) - p(-A))/2$ to make it odd without losing uniform accuracy. Approximate the finitely many coefficients by rationals. Accuracy less than half the original nonvanishing margin preserves nonvanishing.

All surviving monomials have odd total degree. Choose a common odd degree q at least as large as every degree present. Multiply each monomial of degree j by $(\sum A_{uv}^2)^{(q-j)/2}$. This leaves its value unchanged on Z and makes the whole family homogeneous of degree q . Homogeneity extends nonvanishing from Z to the nonzero cone. $\square$

The distinction from RA-17 is not cosmetic. These polynomials need not be linear, and in general $P(X) - P(Y) \neq P(X - Y)$. The kernel/difference argument in Lemma 2.1 cannot be transferred to such a nonlinear family.

4 The evaluation-bundle relaxation has the same critical criterion

Put $B = \text{Gr}_c(\mathbb{R}^d)$, let Q be its tautological rank- c bundle, and put

$$E = dQ, \quad N = dc, \quad b = \dim B = kc, \quad q = c^2 + 1 = N - b + 1.$$

An admissible q -dimensional matrix space would give q independent sections of E by restricting matrices to the planes represented by Q . If a restriction vanished, the matrix would factor through a k -dimensional quotient and have rank at most k .

⁵See source [5].

Theorem 4.1 (Exact critical frame criterion). *The bundle $E = dQ$ has $c^2 + 1$ pointwise independent continuous sections if and only if $D_{d,r}$ is even. Equivalently, in the even-degree case there exists a real bundle η of rank $b - 1$ such that*

$$\varepsilon^{c^2+1} \oplus \eta \cong dQ.$$

Proof. Here $b = kc$ is even and $q \geq 2$. The Stiefel manifold $V_{N,q}$ is $(b - 2)$ -connected, and its first nonzero homotopy group is

$$\pi_{b-1}(V_{N,q}) \cong \mathbb{Z}/2.$$

For completeness, fix the first $q - 2$ frame vectors. The fiber is $V_{b+1,2}$, and the base $V_{N,q-2}$ is b -connected, so the relevant homotopy group is unchanged. The two-frame manifold $V_{b+1,2}$ is the unit tangent sphere bundle of S^b . In its homotopy sequence, the boundary $\pi_b(S^b) \rightarrow \pi_{b-1}(S^{b-1})$ is multiplication by the Euler number $\chi(S^b) = 2$. Its cokernel is $\mathbb{Z}/2$. The case $q = 2$ uses this last calculation directly.

Since the base has dimension b , there is only this first obstruction to a q -frame. It is the class $w_b(E)$, with untwisted mod-two coefficients. There are no higher obstructions on this base. This is the usual obstruction-theoretic definition of the relevant Stiefel–Whitney class.

To evaluate $w_b(E)$, take q generic constant matrices and their restriction sections. A linear dependence at a c -plane means that a nonzero member of the matrix pencil annihilates that plane. Generically such a member has rank exactly k and has a unique c -dimensional nullspace. The degeneracy points therefore correspond exactly to the finite projective intersection with the rank-at-most- k variety. The parity of their count is $D_{d,r}$. Top mod-two cohomology of the connected closed manifold B is $\mathbb{F}_2$, proving that $w_b(E)$ vanishes exactly in the even-degree case. A frame has an orthogonal complement, giving the asserted bundle decomposition. $\square$

Corollary 4.2 (A precise limit of bundle-only arguments). *When $D_{d,r}$ is even, no contradiction to the abstract decomposition*

$$\varepsilon^{c^2+1} \oplus \eta \cong dQ$$

can rule out the critical kernel dimension $c^2 + 1$: such a decomposition actually exists. In particular, characteristic classes or indices derived only from that decomposition cannot exclude it.

This does not say that all topological arguments about the original linear problem are exhausted. An argument using extra compatibility of constant matrix sections, or extra geometry beyond this bundle relation, would be a different argument. It says exactly which relaxation has already become feasible.

5 A direct rank-one index calculation

This section specializes and independently rederives part of the earlier index obstruction. It uses the same kernel-to-bundle principle, not a new construction. Write $d = 2\ell \geq 4$, $r = 1$, and set

$$h = \text{popcount}(\ell - 1), \quad c = d - 2.$$

Theorem 5.1 (Rank-one index bound). *For even $d \geq 4$,*

$$\mu_{\mathbb{R}}(d, 1) \geq 4d - 4 - 2h + (h \bmod 2). \quad (2)$$

In particular, for $d = 2^s + 2$ with $s \geq 1$, the bound is $4d - 5$.

5.1 The relevant characteristic number

Work over $G = \text{Gr}_2(\mathbb{R}^d)$, with tautological S and complementary Q . An admissible kernel of dimension $\kappa = c^2 + j$ forces

$$\varepsilon^\kappa \oplus \eta \cong dQ, \quad t := \text{rank } \eta = 2(d-2) - j.$$

The rational cohomology ring is

$$H^*(G; \mathbb{Q}) = \mathbb{Q}[p]/(p^\ell), \quad |p| = 4, \quad p = p_1(S), \quad \int_G p^{\ell-1} = 1$$

for a suitable orientation. The rational ring is the even/even unoriented Grassmannian presentation.⁶ For the normalization, the oriented two-plane cover identifies with the complex quadric of dimension $d-2$; the Euler class of the oriented two-plane bundle is, up to sign, its hyperplane class. The top hyperplane number of a quadric is two, and the cover has degree two.

Both TG and η are orientable and admit Spin^c structures whose determinant classes are torsion. Indeed, if $u = w_1(S) = w_1(Q)$, then

$$w_2(TG) = (\ell-1)u^2, \quad w_2(\eta) = \ell u^2.$$

The integral Bockstein $\beta(u)$ reduces modulo two to u^2 and is 2-torsion, so it supplies the required integral lifts. Thus the Spin^c index theorem applies, with rational determinant factors equal to one.⁷

Introduce the formal Euler root z of S , so $p = z^2$. Stable identities give

$$TG \oplus (S \otimes S) \cong dS, \quad Q \oplus S \cong \varepsilon^d.$$

The normalized spinor Chern character of η is therefore $\cosh(z/2)^{-d}$, while

$$\hat{A}(TG) = \left(\frac{z/2}{\sinh(z/2)} \right)^d \frac{\sinh z}{z}.$$

Multiplying gives $(z/\sinh z)^{d-1}$. The absolute value of the total spinor index is consequently

$$|I| = 2^{\lfloor t/2 \rfloor - (d-2)} \binom{d-2}{(d-2)/2} = \frac{\binom{d-2}{(d-2)/2}}{2^{\lfloor j/2 \rfloor}}. \quad (3)$$

No assumption that G is spin, rather than merely Spin^c , is needed in this computation.

Here is a direct coefficient verification. Formal change of variables $v = \sinh z$ yields

$$\begin{aligned} [z^{d-2}] \left(\frac{z}{\sinh z} \right)^{d-1} &= \text{Res}_{z=0} \frac{dz}{(\sinh z)^{d-1}} \\ &= [v^{d-2}] (1+v^2)^{-1/2} = (-1)^{(d-2)/2} \frac{\binom{d-2}{(d-2)/2}}{2^{d-2}}. \end{aligned}$$

This derivation avoids an unevaluated localization sum.

⁶See source [3].

⁷See source [7].

5.2 Integrality and the half-spin refinement

The 2-adic valuation of the central binomial coefficient in (3) is $h = \text{popcount}(\ell - 1)$, by the factorial valuation formula. If h is even, taking $j = 2h + 1$ makes the total index a noninteger. Such a kernel is impossible.

If h is odd, instead take $j = 2h$. Then $t \equiv 2 \pmod{4}$. The Euler class of η vanishes rationally because $H^*(G; \mathbb{Q})$ is concentrated in degrees divisible by four. The two half-spinor Chern characters therefore coincide rationally. Each half-index must equal $I/2$, but

$$|I/2| = \frac{\binom{d-2}{(d-2)/2}}{2^{h+1}}$$

is a noninteger. Again the proposed kernel is impossible. A larger admissible kernel would contain the forbidden smaller-dimensional one. The maximum possible excess j is therefore at most $2h - (h \bmod 2)$, proving Theorem 5.1.

5.3 The six-dimensional gap, explicitly

For $d = 6$, $G = \text{Gr}_2(\mathbb{R}^6)$ has

$$H^*(G; \mathbb{Q}) = \mathbb{Q}[p]/(p^3), \quad \int_G p^2 = 1.$$

A hypothetical eighteen-dimensional minimum-rank-three kernel would yield an oriented rank-six η with

$$p_1(\eta) = -6p, \quad p_2(\eta) = 21p^2.$$

One also has $p_1(TG) = 2p$, $p_2(TG) = 7p^2$, and hence

$$\hat{A}(TG) = 1 - \frac{p}{12}.$$

The rational Euler class of η is zero. Its half-spinor character is

$$\text{ch}(\Delta^+ \eta) = 4 - 3p + \frac{5}{4}p^2.$$

The twisted Dirac index would therefore be

$$\int_G \left(1 - \frac{p}{12}\right) \left(4 - 3p + \frac{5}{4}p^2\right) = \frac{3}{2},$$

which is impossible. This proves $\kappa(6, 1) \leq 17$ and $\mu_{\mathbb{R}}(6, 1) \geq 19$.

For a seventeen-dimensional kernel, the complement instead has rank seven. The total spinor character is $8 - 6p + \frac{5}{2}p^2$, giving index 3, an integer. More strongly,

$$D_{6,1} = 1764$$

is even, so Theorem 4.1 proves that a rank-seven complement with $\varepsilon^{17} \oplus \eta \cong 6Q$ actually exists. There is no missing bundle-only nonexistence theorem at this endpoint. What remains missing is a *linear* seventeen-dimensional pencil, or an obstruction using more than this abstract bundle decomposition.

6 Exact algebraic certificates and the interrupted pencil

An algebraic candidate must be distinguished from a verified construction. The accompanying scripts use two exact acceptance mechanisms. Numerical optimization is permitted only as a way of proposing rational data; it is never itself an acceptance test.

6.1 Positive Gram forms in the minor ideal

Suppose $A(x) = \sum_{j=0}^{q-1} x_j B_j$ is a rational pencil, and $f_i(x)$ are all its $(k+1)$ -minors. A certificate consists of rational polynomials h_i , a vector $z(x)$ of all monomials of some positive degree, and a rational symmetric positive-definite matrix H satisfying

$$z(x)^T H z(x) = \sum_i h_i(x) f_i(x). \quad (4)$$

If all $f_i(x)$ vanished at a nonzero real x , the right side would be zero, but the left side would be positive: $z(x) \neq 0$ because it includes pure powers of each coordinate. Thus the pencil is admissible. Identity (4) is checked coefficient by coefficient over $\mathbb{Q}$; positive definiteness is checked by exact rational Schur-complement pivots.

The search script first solves a semidefinite feasibility problem, then rounds free entries to rationals and solves the linear coefficient constraints exactly for the remaining entries. It rejects the proposal unless both exact checks pass. An infeasible numerical solve or a timeout is labelled inconclusive, not a counterexample to the pencil.

6.2 Projective charts and Hermite trace forms

An alternative verification covers projective parameter space by the disjoint charts

$$x_0 = \cdots = x_{i-1} = 0, \quad x_i = 1, \quad i = 0, \dots, q-1.$$

This explicitly includes every projective point, so no hyperplane at infinity is omitted. On a chart with zero-dimensional rational minor ideal I , let $T = \mathbb{Q}[x]/I$ and form the trace pairing

$$H_T(u, v) = \text{Tr}(M_{uv}),$$

where M denotes multiplication on the finite-dimensional algebra. Its signature is the number of distinct real points of the chart. Indeed, each real local factor contributes one positive direction, each conjugate complex pair contributes one positive and one negative direction, and the nilradical contributes only zero directions. Local multiplicities multiply the nonzero trace forms by positive integers and do not change this signature. Thus signature zero proves real emptiness. A unit ideal also proves emptiness; a positive-dimensional complex ideal is reported as inconclusive by this particular checker.

6.3 Status of the preserved small-pencil calculation

No new small-pencil certificate is asserted in this continuation. The earlier Xu certificate remains available in the preserved prior archive.

7 What remains to solve RA-17

The exact linear classification has not been obtained. In particular, neither of the following alternatives has been established for the first retained gap:

$$\begin{aligned} &\text{Construct a seventeen-dimensional } K \subset M_6(\mathbb{R}) \\ &\text{whose every nonzero member has rank at least three,} \\ &\text{or prove that no such linear space exists.} \end{aligned} \tag{5}$$

The first would give $\mu_{\mathbb{R}}(6, 1) = 19$; the second, together with Proposition 2.2, would give $\mu_{\mathbb{R}}(6, 1) = 20$.

Theorems 3.1 and 4.1 solve specific relaxations, not (5). At this endpoint they exclude a proposed strategy: merely strengthening the characteristic-class or index calculation for the same abstract complement cannot prove nonexistence, because the complement exists. They do not exclude a strategy based on linearity, simultaneous compatibility of evaluations, real algebraic geometry, or a direct certified construction.

There are other retained gaps as well. The preceding report records $35 \leq \mu_{\mathbb{R}}(10, 1) \leq 36$, and corank-one intervals at $(32, 15)$ and $(64, 31)$. This continuation does not close those intervals or independently re-prove all of their endpoints. A complete answer requires exact values for every admissible pair, not only a larger finite table or a decision procedure for one fixed pair. No claim of complete solution should be inferred from the existence of a reproducible archive.

A Reproduction, assurance, and source provenance

The primary arithmetic check is

```
python code/verify_relaxation.py
```

It checks the factorial degree formulas, their rank-two binary valuation, the formal-series index coefficient, the indicated mod-two Grassmannian class, and the ranks of the exported integer measurement systems. For the rank-two variety the closed degree formula is

$$D_{d,1} = \frac{(2d-3)!(2d-4)!}{(d-1)!^2(d-2)!^2}, \quad \nu_2(D_{d,1}) = 2(\text{popcount}(d-1) - 1).$$

Thus this degree is odd exactly when $d-1$ is a power of two. Combining with Proposition 2.3 gives $\mu_{\mathbb{R}}(d, 1) = 4d - 4$ for those dimensions. For all other dimensions, this parity computation alone is not an exact linear classification.

The optional pencil checker is run as

```
python code/certify_interrupted_pencil.py --chart 0
# Repeat for charts 1, 2, 3, 4.
python code/sos_pencil_certificate.py --degree 4
# A numerical proposal is not accepted until rational verification passes.
```

Saved Gram certificates can be verified without re-running the SDP by supplying `--verify` and the certificate filename. Such verification does not need CVXPY. The chart method recomputes the rational Groebner basis rather than assuming that an arbitrary stored basis generates the original minor ideal.

The files under `logs/` record the actual executions and their exit statuses. A missing certificate, timeout, or positive-dimensional chart is not labelled a proof of admissibility. Arithmetic checks

certify identities and matrix computations, not the correctness of every topological inference. The written proofs are not proof-assistant formalizations and have not been independently peer-reviewed as part of this archive.

The earlier archives are included under `prior/` without modification. The copied interrupted scripts under `exploratory/` are research notes, not additional theorems. Third-party research PDFs are not redistributed. A SHA-256 manifest records the delivered files; archive integrity is distinct from mathematical correctness.

References

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